

✓ Power Of Four

Question

Given an integer n , return true if it is a power of four. Otherwise, return false.

An integer n is a power of four, if there exists an integer x such that $n == 4^x$.

Example 1:

- Input: $n = 16$
 - Output: true
-

Constraints

- $-2^{31} \leq n \leq 2^{31} - 1$
-

Solution

so for this question here i do it by using recursion..

Code

```
class Solution {
    public boolean helper(int num) {
        if (num == 0 ){
            return false;
        }
        if (num == 1) {
            return true;
        } else {
            if (num % 4 != 0) {
                return false;
            }
            return helper(num / 4);
        }
    }

    public boolean isPowerOfFour(int n) {
        if (n == 1) {
```

```

        return true;
    }
    if (n % 4 != 0) {
        return false;
    } else {
        return helper(n);
    }
}
}

```

✓ dry run of above code

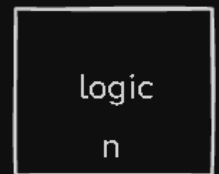
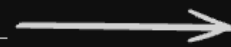
let run this code for 0 to 10 and examine the outputs.

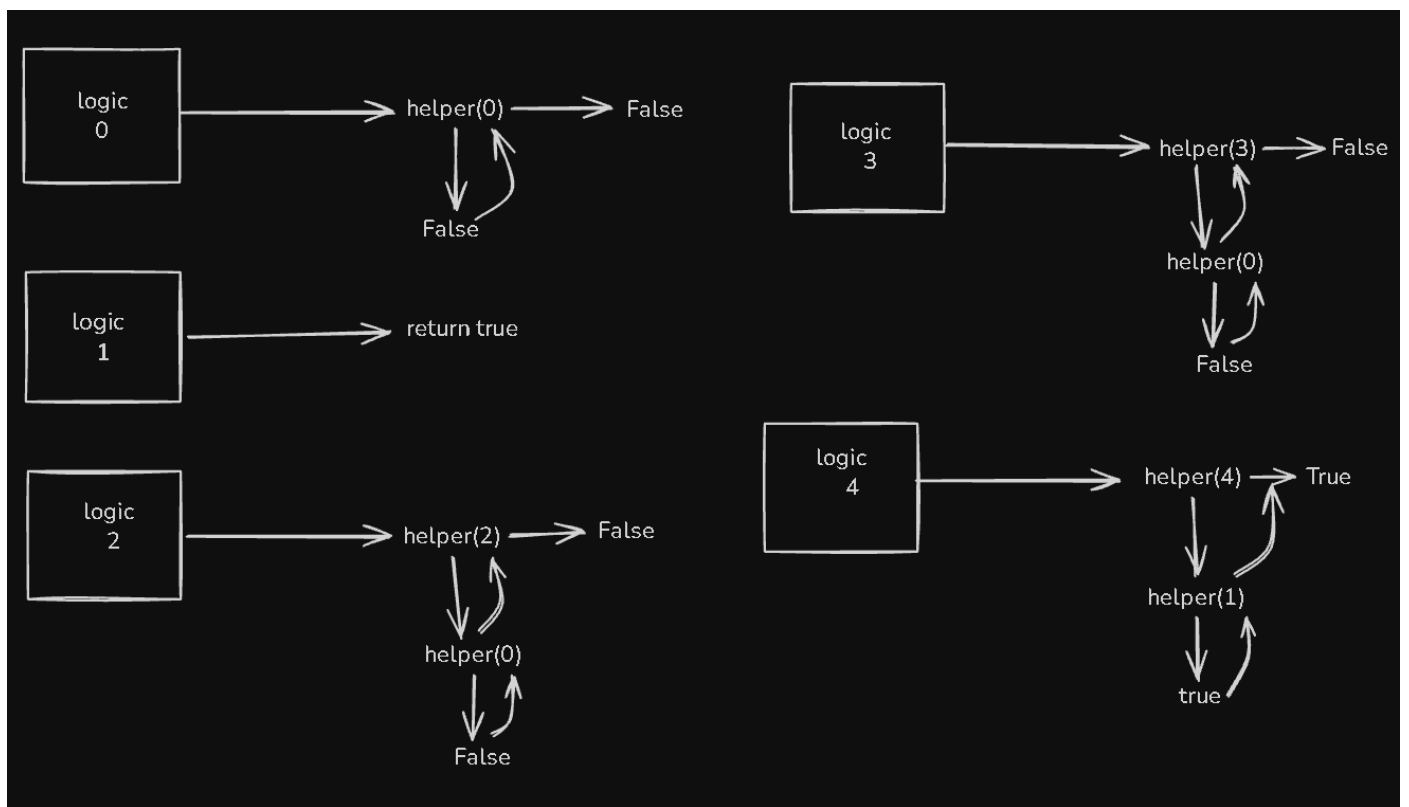
input	output
0	False
1	True
2	False
3	False
4	True
5	False
6	False
7	False
8	False
9	False
10	False

```

isPowerOfFour(int n){
    _____
    _____
    _____
}

```





time complexity = $O(\log(n))$

space complexity = $O(1)$